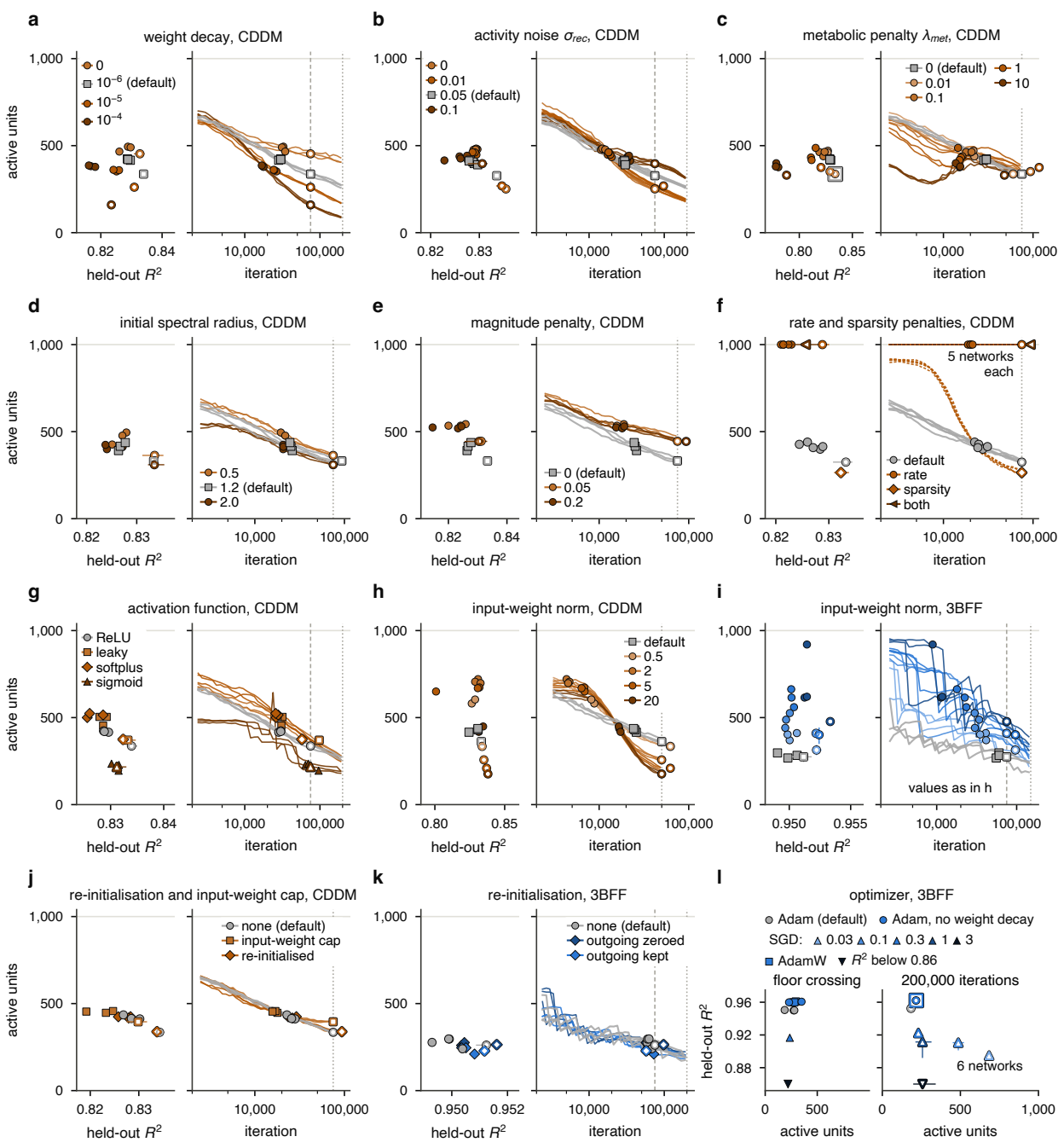

a
weight decay, CDDM
0
10^{-6} (default)
10^{-5}
10^{-4}
b
activity noise σ_{rec}, CDDM
0
0.01
0.05 (default)
0.1
c
metabolic penalty λ_{met}, CDDM
0 (default)
0.01
0.1
1
10
d
initial spectral radius, CDDM
0.5
1.2 (default)
2.0
e
magnitude penalty, CDDM
0 (default)
0.05
0.2
f
rate and sparsity penalties, CDDM
5 networks each
default
rate
sparsity
both
g
activation function, CDDM
ReLU
leaky
softplus
sigmoid
h
input-weight norm, CDDM
default
0.5
2
5
20
i
input-weight norm, 3BFF
values as in h
j
re-initialisation and input-weight cap, CDDM
none (default)
input-weight cap
re-initialised
k
re-initialisation, 3BFF
none (default)
outgoing zeroed
outgoing kept
l
optimizer, 3BFF
Adam (default)
Adam, no weight decay
SGD: 0.03 0.1 0.3 1 3
AdamW
R^2 below 0.86
floor crossing
200,000 iterations
6 networks
active units
held-out R^2
iteration